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Inconsistency in Neuroscience

- 1 Terminology is used inconsistently across studies to describe the same functional unit. Alternatively, the same term may describe different entities.
- 2 Differences in methodology, datasets, and individual subject variability complicate things further (see Figure 1).
- 3 Inconsistency causes confusion and impedes scientific progress. Standardized functional units and nomenclature are needed.

ICN Spatial Maps

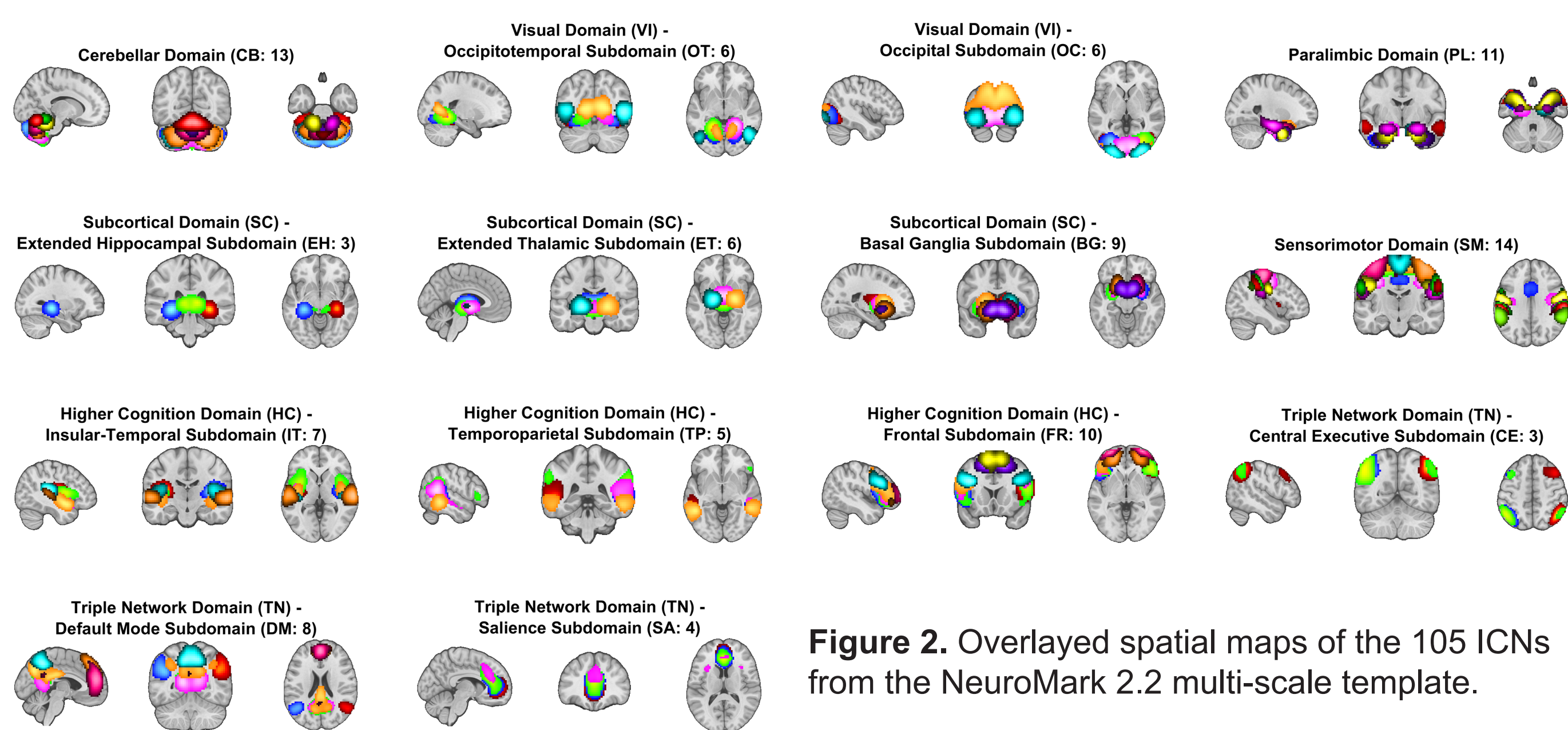


Figure 2. Overlaid spatial maps of the 105 ICNs from the NeuroMark 2.2 multi-scale template.

Why use the NeuroMark 2.2 Atlas?

- 1 It is highly replicable, well-equipped for comparing findings across studies, and is sensitive to individual differences.
- 2 It encompasses the whole-brain, incorporates information from multiple spatial scales, and has demonstrated reliability across the lifespan.
- 3 It describes functional units in terms familiar to the fields of cognitive and affective neuroscience to improve harmonization and standardization.

Triple Network Domain (TN)

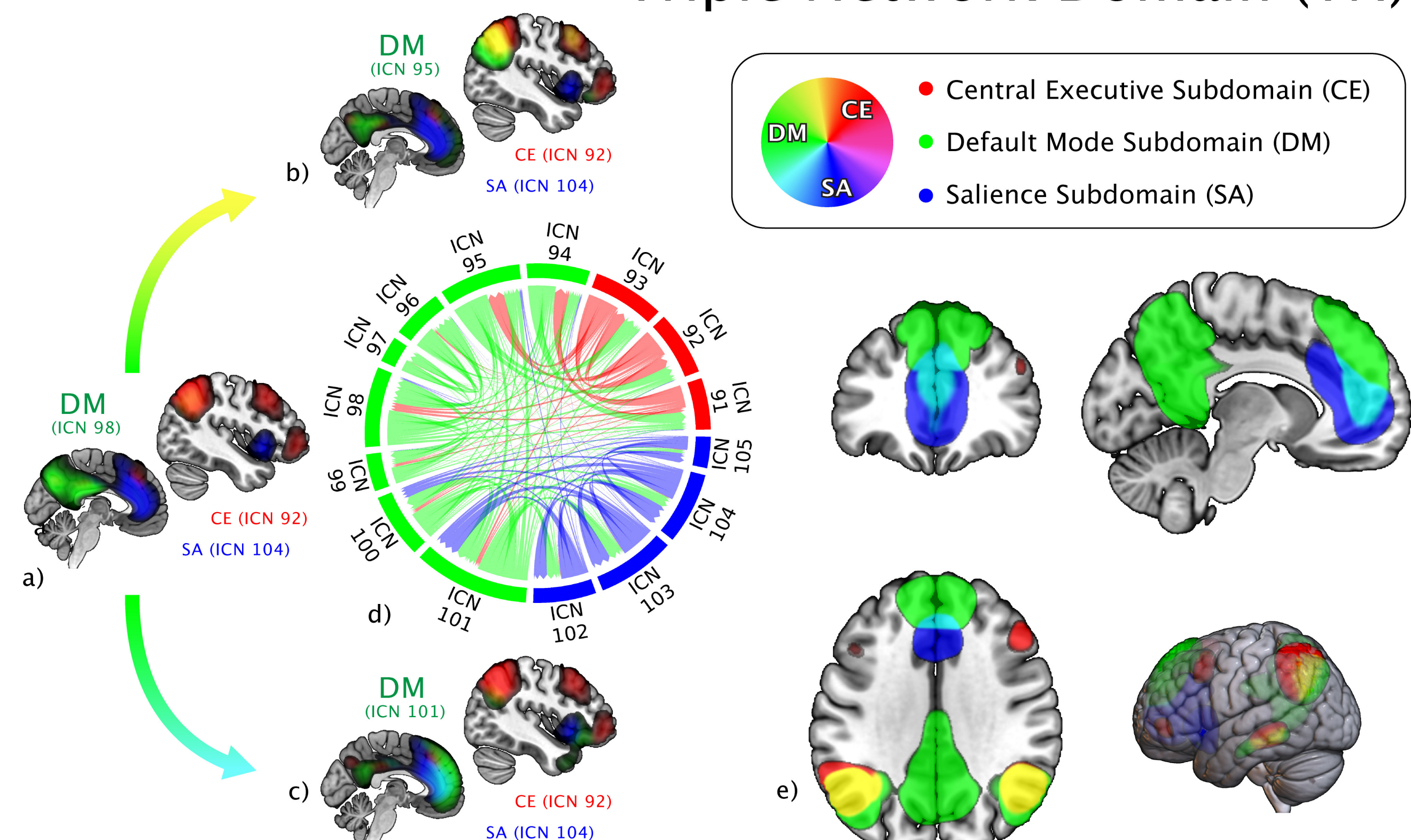


Figure 4. Spatial maps are overlaid, revealing an apparent transition where a) DM ICN 98 displays relatively little overlap with CE and SA ICNs, b) DM ICN 95 displays greater overlap with CE ICN 92, and c) DM ICN 101 displays greater overlap with SA ICN 104. d) The proportion of significant voxels (z-score > 1.96) overlapping between all TN ICNs. e) Spatial maps for all TN ICNs are overlaid on the brain.

Individual Variability of a Functional Unit

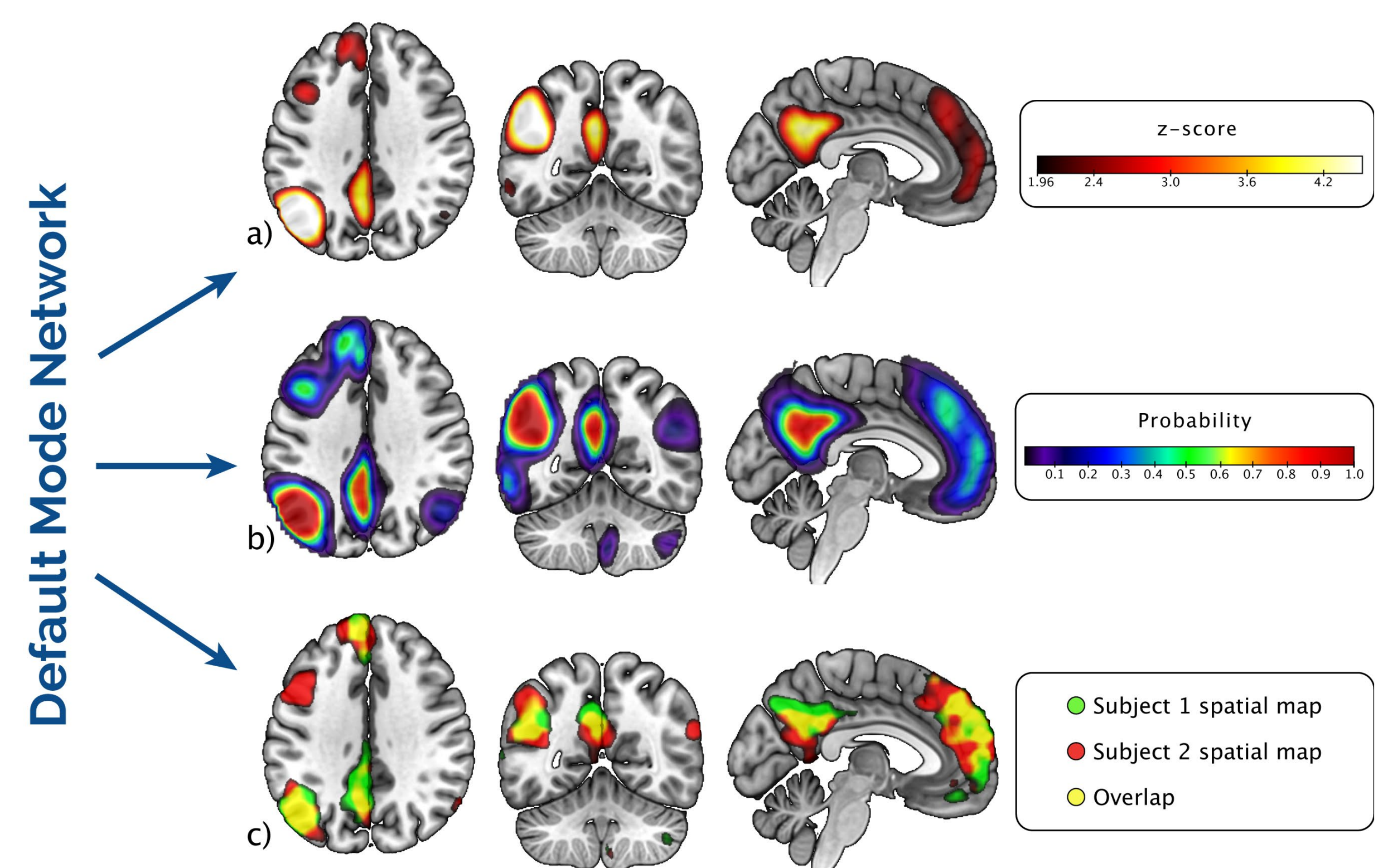
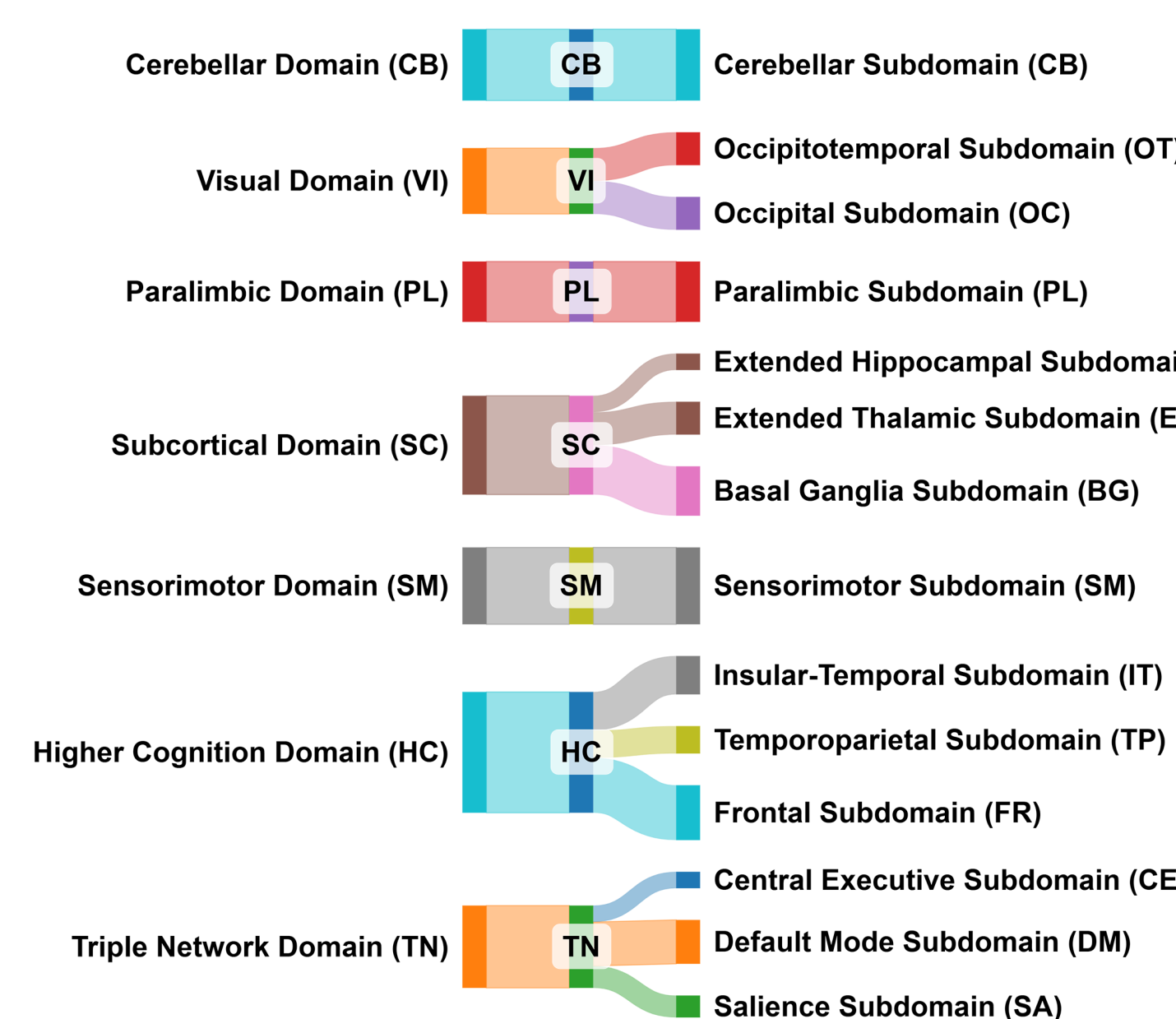


Figure 1. a) The average of ~40k individual subject spatial maps for an intrinsic connectivity network (ICN). Voxels represent significant z-scores (> 1.96). b) The probability that a given voxel of the ICN is significant. c) Spatial maps for two subjects demonstrate a great amount of individual variability.

What is NeuroMark?

- 1 NeuroMark represents efforts towards establishing a common reference space across functional imaging analyses.
- 2 Past efforts include a standardized fMRI preprocessing and analysis pipeline which incorporates a template (Fig. 2) in spatially-constrained ICA.
- 3 NeuroMark has been utilized to identify imaging markers associated with various clinical groups.

Domain - Subdomain



Average FNC

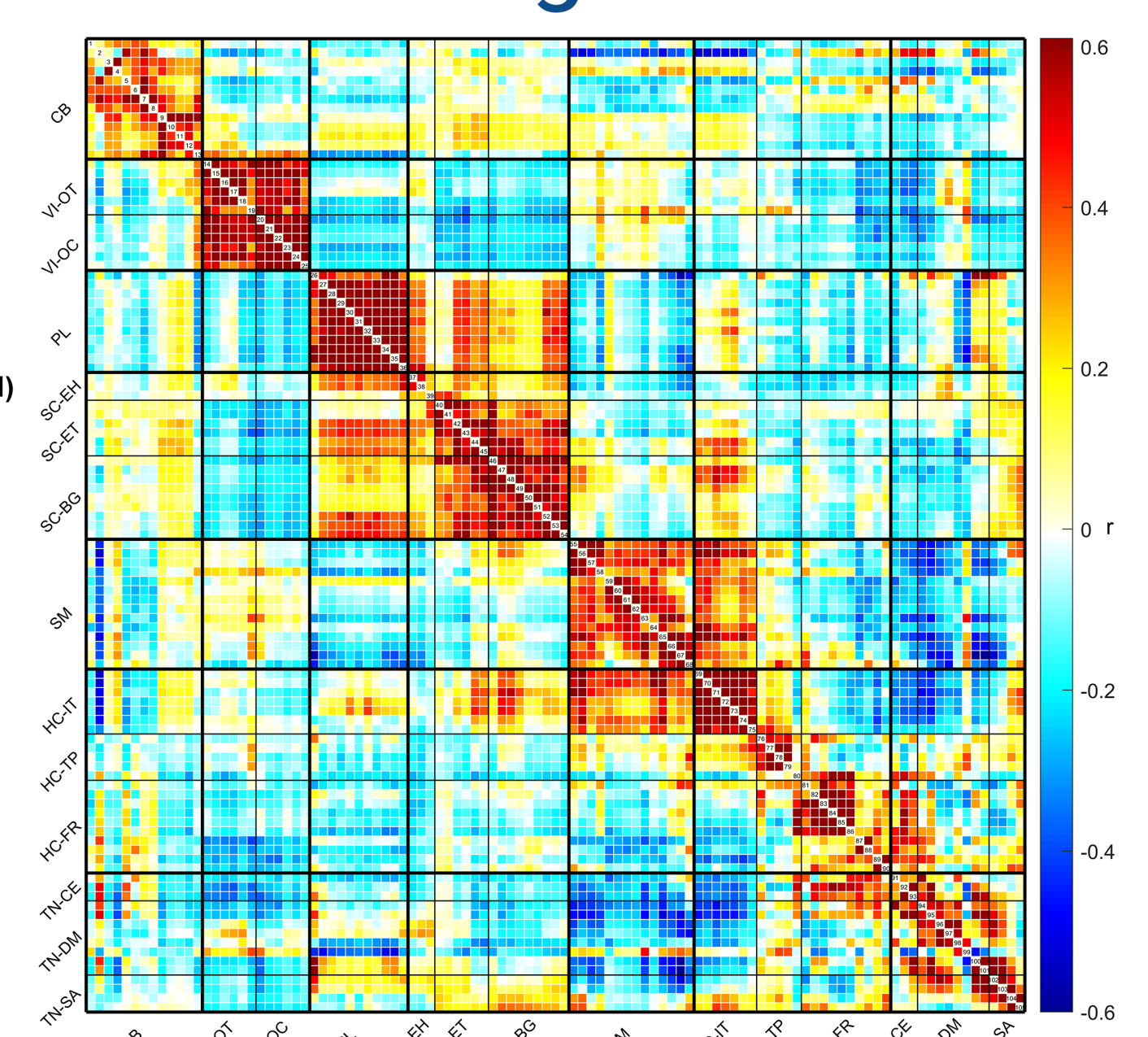


Figure 3. Functional network connectivity (FNC) calculated between 105 ICNs from the NeuroMark 2.2 template is averaged across ~40k participants from the UKB dataset and grouped by domain-subdomain.

The Integrative Functional Brain

- 1 The atlas reveals how brain networks may be more fluid and integrative than how they are traditionally conceptualized.
- 2 Despite their incorporation of spatially distinct brain regions, ICNs are highly modular both within and across domains (Fig. 3), demonstrating heterogeneity of function across brain regions as well as the recruitment of multiple structures to perform functions.
- 3 The spatial gradient (Figure 4) in the Triple Network domain suggests that the Central Executive, Default Mode, and Salience networks may be better described as a single continuum.

